

Education

University of Colorado at Boulder	May 2026
B.S., Aerospace Engineering GPA: 3.476	Boulder, CO
Relevant Coursework: Medicine in Space Environments; Orbital Mechanics; Aircraft Dynamics; Dynamics Controls Lab; Electronics; Thermodynamics and Heat Transfer; Structures; Aerodynamics; Foundations of Propulsion	
Computer Science Minor	
- C/C++, MATLAB, and Python programming with low-level Assembly processes - Coursework: Algorithms; Data Structures; Computer Systems; Database Systems; Computational Methods	

Skills

Programming: C, C++, Python, MATLAB, SQL
Software: Adobe Photoshop/Lightroom, GitHub, SolidWorks, OpenVSP, EAGLE, LaTeX/Overleaf, MATLAB Simulink, CMake
Hardware: Rapid Prototyping, 3D Printing, PCB Manufacturing, Woodworking, Machine Shop, Soldering

Work Experience

Raytheon	August 2026 – Present
Effectors Software Engineer I	Tucson, AZ
- Manage repository branches and software compilation processes utilizing CMake and GitHub through Linux environments. - Collaborate on high-level system processes and software development in C++.	
Epic Productions	May 2024 – August 2025
Lead Wedding Photographer	Boulder, CO
- Managed wedding projects through planning and execution, ensuring client requirements and deadlines were met. - Used Adobe and Microsoft tools for time management and project planning.	
City of Austin	May 2021 – Aug 2023
Head Lifeguard	Austin, TX
- Maintained emergency procedures, performed risk assessments, and managed scheduling. - Instructed swim lessons for children and coached the youth swim team.	

Technical Experience

- Retractable Canard Passenger Aircraft: - Programmed INAV firmware on a fixed-wing prototype for flight control, telemetry, and video. - Prototyped a retractable canard on a sub-scale technological demonstrator to analyze transient stability behavior. - Performed control analysis in MATLAB to determine control surface sizing and modal behavior. - Presented a paper at the AIAA student conference and showcased work to professors in design review presentations. - EVA Medical Decision Aid (Mars Desert Research Station) - Validated the aid in an analog environment at the MDRS to evaluate performance in simulated conditions. - Developed a risk decision matrix by organizing NASA medical standards into a weighted model to quantify likelihoods. - Engineered an interface to reduce cognitive load, enabling mechanism of injury (MOI) triage and critical decision-making during EVAs. - Published the development and testing of the decision aid for presentation at an industry conference. - Arthritis Assisting Glove: - Designed and manufactured a custom PCB using EAGLE CAD. - Integrated flex sensors with an Arduino in a wearable glove for assisted finger movement. - Boost-Glide Project: - Modeled an air-propelled bottle rocket using predictive simulation in MATLAB. - Iteratively fabricated the design in a machine shop and performed a boost-glide test.
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Leadership and Activities

AIAA	Nov 2025 – May 2026
Officer	Boulder, CO
- Organized campus networking events between students and the Aerospace industry. - Represented CU at conventions and industry events.	
Westlake Video Crew	Aug 2019 – Dec 2021
Crew Head	Austin, TX
- Served as Stage Manager for a crew of 40+ members over 6 sold-out technical productions. - Conducted safety briefings and coordinated a full orchestra, a dozen sub-crews, and first responders.	

Achievements

- 1st Place Freshman Projects Showcase: Arthritis Assisting Glove
- 1st Place UIL Broadcasting 2018: Westlake Video Crew